

The propositions package*

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Abstract

The `propositions` package provides a key-value driven system for labelling propositions, theses, and premises in academic papers. Items may be given names like ‘(P)’ or ‘Physicalism’, or auto-numbered using different counters; all carry robust cross-references with configurable formatting. The package integrates with `amsmath`, `hyperref`, and `cleveref`.

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1 Introduction

In some academic disciplines (such as philosophy), it is common to have displayed propositions (examples, theses, premises, . . .) with various kinds of labels. A thesis might be referred to as ‘(P)’ or ‘Physicalism’; the premises of an argument might be numbered as ‘P1’, ‘P2’, ‘P3’, . . . ; or examples might be numbered consecutively over the course of a whole article. Standard L^AT_EX environments like `enumerate` can handle some of these cases, but cross-referencing is awkward: `\ref` produces a bare number or letter, and the author must manually add parentheses or other formatting at every point of reference. The standard `description` environment does not allow cross-referencing at all.

The `propositions` package solves this by attaching formatting information to each label. A short item like `\pitem[P]` is displayed as “(P)” and `\ref` automatically produces “(P)” as well—complete with parentheses, hyperlinks, and `cleveref` support. The full key-value interface supports named items, numbered items, custom counters, glosses, shorthands, and per-item format overrides.

The `\ptag` command (which requires `amsmath`) extends this to displayed math environments: an equation can be tagged with a proposition label instead of (or using) its equation number.

2 History

I wrote the ancestor to this package in the 90s while finishing my Ph.D. thesis, but never documented it or shared it with the world. This new version is a thorough re-implementation in L^AT_EX3, written in 2026 with extensive help from Claude Code. I hope others will find it as useful as I have.

3 Basic usage

Load the package with `\usepackage{propositions}` or `\usepackage[options]{propositions}` (see [section 8](#) below for valid package options).

The `prop` environment generates a list of propositions, each introduced by a `\pitem`. `\pitem` with an optional argument gives a `description`-like label:

<code>\begin{prop}</code>	
<code>\pitem[Physicalism] Everything is</code>	Physicalism Everything is physi-
<code>physical. \label{phys}</code>	cal.
<code>\pitem[Idealism] Everything is</code>	Idealism Everything is mental.
<code>mental. \label{ideal}</code>	
<code>\end{prop}</code>	

Unlike the standard `description` environment, one can refer back to these propositions using the standard `\ref` command:

<code>\ref{phys}</code> is more plausible	Physicalism is more plausible than
than <code>\ref{ideal}</code> .	Idealism .

With no optional argument, `\pitem` will by default generate numbered items similar to `enumerate`, but with numbering that persists across the document:

<pre> \begin{prop} \pitem Every atom is physical. \label{atoms} \end{prop} \ref{phys} follows from the conjunction of \ref{atoms} and \begin{prop} \pitem Everything is an atom. \label{atomism} \end{prop} </pre>	(1) Every atom is physical. Physicalism follows from the conjunction of (1) and (2) Everything is an atom.
--	--

As with `enumerate`, the counter and formatting depend on the nesting level:

<pre> \begin{prop} \pitem \label{dual} \begin{prop} \pitem Some things are physical. \label{some} \pitem Some things are not physical. \label{notall} \end{prop} \end{prop} Without \ref{some}, \ref{dual} would be consistent with \ref{ideal}. </pre>	(3) a. Some things are physical. b. Some things are not physical. Without (3a), (3) would be consistent with Idealism.
---	--

4 Advanced usage

The format of the proposition labels, and of subsequent references, are both configurable using a key=value syntax (see section 5 for the possible keys):

<pre> \begin{prop} \pitem[No Overlap, align=flush, display format=\textbf{#1}, ref format=\textit{#1}] Nothing mental is physical. \label{incomp} \end{prop} Is \ref{dual} consistent with \ref{phys}? </pre>	No Overlap Nothing mental is physical. Is (3) consistent with Physicalism?
---	--

Preset *item types* can be declared and used instead of configuring the `\pitems` one at a time:

<pre> \begin{prop} \pitem[Nihilism, type=long] There is nothing. \label{nihilism} \end{prop} Does \ref{nihilism} imply \ref{phys}, \ref{dual}, or both? Discuss. </pre>	Nihilism There is nothing. Does Nihilism imply Physicalism, (3), or both? Discuss.
---	---

Shortcut commands—e.g., `\litem[⟨keys⟩]` for `\pitem[type=long, ⟨keys⟩]`—can also be defined. The package loads with a range of predefined types.

When the package is loaded with `\usepackage[equations]{propositions}`, first-level `\pitems` use the same counter as equations. (This looks better with the `leqno` option to `\documentclass`.)

<pre>\begin{equation} \exists x (\text{Mental}(x) \wedge \text{Physical}(x)) \end{equation} \begin{prop} \pitem There is overlap between the mental and the physical. \end{prop}</pre>	$(4) \quad \exists x(\text{Mental}(x) \wedge \text{Physical}(x))$ (5) There is overlap between the mental and the physical.
--	--

The `\ptag` command (requires `amsmath`) is an analogue of `\pitem` that works inside displayed math environments.

<pre>\begin{equation} \ptag[Monism] \label{mon} \exists x \forall y (y = x) \end{equation} Is \ref{mon} compatible with \ref{dual}?</pre>	$\text{Monism} \quad \exists x \forall y (y = x)$ Is Monism compatible with (3)?
---	--

5 The prop environment and \pitem

```
\begin{prop}[\langle keys \rangle]
  \langle environment content \rangle
\end{prop}
```

Creates a displayed list of propositions. It is a standard L^AT_EX list, so by default its formatting will depend on the standard length parameters like `\itemsep` and `\topsep`, although these can be overridden by setting package keys.

Within `prop`, `\pitem` (see below) creates labelled items. The ordinary `\item` command is still available for unlabelled items.

Any display math environment (`\equation`, `align`, etc.) used inside `prop` or `inlineprop` automatically increments the nesting level for its duration, so that `\ptag` inside such an environment behaves as if it were one level deeper. The optional `\langle keys \rangle` argument accepts the same keys as `\proptoptions`^{→ P. 10} (section 8), with effects local to this environment.

```
\begin{inlineprop}[\langle keys \rangle]
  \langle environment content \rangle
\end{inlineprop}
```

Like `prop`, but does not create a list. Allows `\pitem` to be used outside list environments, e.g. for generating numbers at the beginning of paragraphs. Steps the `prop` counter and increments the nesting level. Accepts the same optional `\langle keys \rangle` as `prop`.

```
\pitem[\langle keys \rangle]
```

Inside the `prop` and `inlineprop` environments, introduces a labelled proposition. The optional argument is a comma-separated list of `\langle key \rangle = \langle value \rangle` pairs.

When used without an optional argument (or without setting `name`, `counter`, or `type`), it behaves as `\pitem[type=<type>]`, where `<type>` depends on the nesting depth. The defaults are `numbered`, `leveltwo`, `levelthree`, `levelfour`, `levelfive`; these can be changed with the `level <n>` keys (section 8).

The following keys can be used in the optional argument of `\pitem`:

`name=<text>` (no default)

The proposition's name. A bare string (without `=`) in the key list is equivalent to `name=<text>`.

`type=<type>` (no default)

An item type, equivalent to a preset collection of keys. Types can be declared using `\SetItemType`^{P.6} or `\DeclareNumberedType`^{P.7}, and several come predefined (section 6).

`counter=<name>` (no default)

Counter to use. The counter is automatically stepped, and the item's `name` is set to `\the<name>`, though this can be overridden by explicitly setting `name`.

`align=<type>` (no default)

How the label should be positioned. Possible values: `default` (offset controlled by `\labelwidth` and `\labelsep`), `right` (right-aligned, like `enumerate`), `center` (centered within the label box), `left` (aligned with left margin of surrounding text), `flush` (aligned with left margin of item text), `nextline` (label on its own line), `left-nextline`, and `flush-nextline`. Has no effect inside `inlineprop` or `\ptag`.

`format=<template>` (no default)

Formatting applied to the `name`: use `#1` for the argument, e.g. `format=\textbf{(#1)}`. Shorthand for setting both `display format` and `ref format`.

`name format=<template>` (no default)

Format for displaying the name in the proposition's label. Does not affect cross-references. (Synonym: `display format`.)

`display format=<template>` (no default)

Synonym for `name format`.

`label format=<template>` (no default)

Format applied to the *entire* assembled label, i.e. the name (after `name format`), shorthand, and gloss together. Use `#1` for the whole assembled text. For example, `label format=#1`: appends a colon after the complete label. Does not affect cross-references.

`ref format=<template>` (no default)

Format for subsequent cross-references to this proposition. Does not affect the display.

shorthand= $\langle text \rangle$ (no default)

An abbreviation displayed after the name. If present, the shorthand becomes the reference text: `\ref` produces the shorthand (formatted with `ref format`) rather than the full name.

shorthand format= $\langle template \rangle$ (initially $\sim$ [#1])

Format for displaying the shorthand in the label.

gloss= $\langle text \rangle$ (no default)

A parenthetical gloss displayed after the name. Does not affect cross-references.

gloss format= $\langle template \rangle$ (initially $\sim$ (#1))

Format for displaying the gloss in the label.

ref= $\langle text \rangle$ (no default)

Explicitly set the reference text, overriding what would be derived from `name`, `counter`, or `shorthand`.

label= $\langle label \rangle$ (no default)

Equivalent to a trailing `\label{\langle label \rangle}`.

crefname= $\langle type \rangle$ (no default)

When `cleveref` is loaded, assigns an arbitrary reference type to this proposition. For example, `crefname=lemma` causes `\cref` to use the names defined by `\crefname{lemma}{...}{...}` instead of the default `proposition` type. The $\langle type \rangle$ must be known to `cleveref`; new types can be declared with `\crefname`.

\ptag[$\langle keys \rangle$]

Available only when `amsmath` is loaded. Works inside displayed math environments like `equation` and `align`. Accepts the same keys as `\pitem`^{P.4}, except that `align` has no effect (positioning is controlled by the tag placement system).

6 Item types

\SetItemType{ $\langle name \rangle$ }{ $\langle keys \rangle$ }

Defines or modifies an item type for use with the `type` key. All `\pitem`^{P.4} keys are accepted, plus the following:

macro= $\langle command \rangle$ (no default)

A new user macro, equivalent to `\pitem[type= $\langle name \rangle$ ]`. Any further keys given to the macro are passed to `\pitem`.

If the type $\langle name \rangle$ already exists, `\SetItemType` modifies or adds keys. For example, `\SetItemType{default}{align=flush}` changes the alignment of the built-in `default` type while preserving its other settings.

<pre> \SetItemType{angle}{ align = left, display format = \textbf{\$\langle\angle\#\!1\angle\rangle\$}, ref format = \$\langle\angle\#\!1\angle\rangle\$, macro = \angitem } \begin{prop} \angitem[Angle thesis] Everything is angular. \end{prop} No further discussion of \lastref{} is needed. </pre>	⟨Angle thesis⟩ Everything is angular. No further discussion of ⟨Angle thesis⟩ is needed.
--	---

⟨DeclareNumberedType{⟨name⟩}[⟨keys⟩]

Creates a new L^AT_EX counter named *⟨name⟩* and a matching item type with *counter=⟨name⟩*. All `\SetItemType`^{→ P.6} keys are accepted, plus:

parent=⟨counter⟩ (no default)

A parent counter; the new counter resets when the parent steps (same mechanism as `\numberwithin`). A dedicated **prop** counter (stepped by each **prop** and **inlineprop**) is available for non-persistent numbering.

counter format=⟨format⟩ (default *⟨name⟩\arabic{⟨name⟩}*)

The representation of the new counter (`\the⟨name⟩`).

<pre> \DeclareNumberedType{P} \begin{prop} \pitem[counter=P] First premise. \label{p1} \pitem[counter=P] Second premise. \label{p2} \end{prop} From \ref{p1} and \ref{p2}\ldots </pre>	P1 First premise. P2 Second premise. From P1 and P2...
--	---

6.1 Built-in types

The following item types are predefined. `\SetItemType`^{→ P.6} can modify their behaviour.

Type	Shortcut	Counter	Display	Ref	Align
plain	none	none	Name	Name	left
default	none*	none	Name	Name	left
long	\litem	none	Name	Name	flush
bullet	\bitem	none	•	•	default
roman	\ritem	roman	(i)	(i)	left
alph	\aitem	alph	(a)	(a)	left
numbered	none [†]	numprop ^{i†}	(1)	(1)	left
leveltwo	none [†]	numprop ⁱⁱ	a.	(1a)	left
levelthree	none [†]	numprop ⁱⁱⁱ	(i)	(1a.i)	left
levelfour	none [†]	numprop ^{iv}	•	•	default
levelfive	none [†]	numprop ^v	–	–	default

- * The `default` type is auto-selected when `\pitem` or `\ptag` has an optional argument but no `type` key.
- † The `numbered-levelfive` types are auto-selected when `\pitem` or `\ptag` has no optional argument, depending on nesting level.
- ‡ The `equations` package option changes `numbered`’s counter to `equation`. The `numpropii`–`numpropv` counters reset automatically when a `\pitem` at the next lower level is processed.

7 Cross-referencing

Labels placed after `\pitem` items work with the standard `\label`/`\ref` mechanism. The key difference from ordinary L^AT_EX references is that `\ref` produces *formatted* output: for example, `\textbf` might be applied to the name, or the number might be wrapped in parentheses. The formatting is controlled by the `format` key (or separately by `display format` and `ref format`).

`\Ref{<label>}`
`\Ref*{<label>}`

Titlecase variant of `\ref`: uppercases the first letter of the formatted output. For example, if `\ref{thesis}` produces ‘the Identity Theory’, then `\Ref{thesis}` produces ‘The Identity Theory’. The starred form suppresses the hyperlink. Note: `\Ref` only affects propositions references (those stored via `\propapply`); for other label types, it behaves like `\ref`.

`\nref{<label>}`
`\nref*{<label>}`
`\Nref{<label>}`
`\Nref*{<label>}`

“Naked ref.” Outputs the bare reference content with all formatting stripped. If `\ref{premise}` produces ‘(P1)’, then `\nref{premise}` produces ‘P1’. The starred form suppresses the hyperlink. `\Nref` is the titlecase variant: it uppercases the first letter of the bare content.

`\nref` is often useful in the argument of `\pitem`, when the the name of one proposition should depend on that of another:

<code>\SetItemType{default}{format=(#1)}</code>	(Phys) Everything is physical.
<code>\begin{prop}</code>	(7*) Almost everything is physical.
<code>\pitem[Phys]</code>	
Everything is physical.	
<code>\label{premise}</code>	(7*) The version (7*), which uses
<code>\pitem[\nref{premise}*]</code>	<code>\nref</code> , looks better than the one
Almost everything is physical.	with <code>\ref</code> , unless for some reason
<code>\label{newpremise}</code>	one wants two lots of parentheses.
<code>\pitem[\ref{premise}*]</code>	
The version <code>\ref{newpremise}</code> , which	
uses <code> \nref </code> , looks better	
than the one with <code> \ref </code> , unless	
for some reason one wants two	
lots of parentheses.	
<code>\end{prop}</code>	

Warning: documents where the name of one item includes a reference to that of another, and there are further references to that item, will require multiple L^AT_EX runs to resolve all references. To save time, it is better to avoid long chains of dependencies of this sort.

```
\oref[⟨prefix⟩][⟨suffix⟩]{⟨label⟩}
\oref*[⟨prefix⟩][⟨suffix⟩]{⟨label⟩}
\Oref[⟨prefix⟩][⟨suffix⟩]{⟨label⟩}
\Oref*[⟨prefix⟩][⟨suffix⟩]{⟨label⟩}
```

“Ref with options.” Extends `\ref` by injecting a prefix and/or suffix *inside* the formatting. With one optional argument, `⟨suffix⟩` is appended; with two, `⟨prefix⟩` is prepended and `⟨suffix⟩` appended. For instance, if `\ref{premise}` produces ‘(P1)’, then `\oref[*]{premise}` produces ‘(P1*)’ and `\oref[cf.~]{*}{premise}` produces ‘(cf.~P1*)’. The starred form suppresses the hyperlink. `\Oref` is the titlecase variant: it uppercases the first letter of the output.

`\oref` can also be useful in the name of `\pitem`, if one wants the display format for the modified item to depend on that originally used

<pre>\begin{prop} \label{premise} \pitem[name=\oref[*]{premise}, format=#1] This will use boldface and parentheses because the original referenced item did. \end{prop}</pre>	7 This will use boldface and parentheses because the original referenced item did.
---	---

Another handy use for `\oref` is in combination with `\nref` to refer to ranges:

<pre>The first two numbered examples in this document were were \oref[--\nref{atomism}]{atoms}.</pre>	The first two numbered examples in this document were were (1–2).
---	--

```
\lastref[⟨prefix⟩]{⟨suffix⟩}
\Lastref[⟨prefix⟩]{⟨suffix⟩}
```

Formatted reference to the most recently processed `\pitem` or `\ptag`, even without a `\label`. Useful for back-references in running text. With one argument, `⟨suffix⟩` is appended; with two, `⟨prefix⟩` is also prepended. Use `\lastref{}` for a plain reference. `\Lastref` is the titlecase variant: it uppercases the first letter.

`\nLastref`

Like `\lastref{}` but returns the bare content without formatting. Takes no arguments; simply output any desired suffix directly afterwards.

```
\parentref[⟨prefix⟩]{⟨suffix⟩}
\Parentref[⟨prefix⟩]{⟨suffix⟩}
```

Inside a nested `prop` (or `inlineprop`), produces a formatted reference to the most recent item of the enclosing level. Same argument convention as `\lastref`. `\Parentref` is the titlecase variant: it uppercases the first letter.

`\nParentref`

Like `\parentref{}` but returns the bare content without formatting. Takes no arguments; simply output any desired suffix directly afterwards.

`\parentref→P.9` and `\nParentref→P.9` are useful for making subitems whose names derive from their parent's:

<pre>\DeclareNumberedType{inner}[counter format=\alph{inner}, display format=#1.] \begin{prop} \pitem[OI] Outer item. \label{outer2} \begin{prop} \pitem[type=inner, format=\parentref{.#1}] \label{dsub1} Ref: \ref{dsub1}, naked: \nref{dsub1}. \pitem[type=inner, display format=#1., ref format=\parentref{.#1}] \label{dsub2} Ref: \ref{dsub2}, naked: \nref{dsub2}. \end{prop} \end{prop}</pre>	(OI) Outer item. (OI.a) Ref: (OI.a), naked: a. b. Ref: (OI.b), naked: b.
--	---

The built-in `leveltwo` and `levelthree` types have `ref format=\parentref{#1}` and `ref format=\parentref{.#1}`, respectively, so that if the parent references as ‘(P1)’, a `leveltwo` sub-item references as ‘(P1a)’ and `\nref` returns just ‘a’.

7.1 How it works: `\propapply`

`\propapply{<template>}{<content>}`

Internally, each reference is stored in the `.aux` file as `\propapply{<template>}{<content>}`.

The `<template>` contains formatting with the placeholder `\propfmtarg` where content appears. At reference time, `\propapply` evaluates the template with `\propfmtarg` bound to `<content>`. The `\oref→P.8` and `\nref→P.8` commands work by locally redefining `\propapply`.

In normal use, you need not interact with `\propapply` directly.

`\propfmtarg`

Placeholder used inside templates; expands to the content argument of the enclosing `\propapply`.

8 Package options

`\proptions{<keys>}`

Sets package-level keys. These can also be set:

- In the optional argument of `prop` and `inlineprop` (local to that environment).
- In the preamble with `\usepackage[⟨options⟩]{propositions}` (global).

Exception 1: keys containing # (such as `equation format`) cannot be set in the optional argument of `\usepackage`, due to how L^AT_EX handles # in option values.

Exception 2: `equations` is a global key; it cannot be used in the optional argument of `prop` or `inlineprop`.

8.1 List dimensions

`topsep`=⟨length / length list⟩
`partopsep`=⟨length / length list⟩
`itemsep`=⟨length / length list⟩
`parsep`=⟨length / length list⟩
`leftmargin`=⟨length / length list⟩
`rightmargin`=⟨length / length list⟩
`labelwidth`=⟨length / length list⟩
`labelsep`=⟨length / length list⟩
`itemindent`=⟨length / length list⟩
`listparindent`=⟨length / length list⟩

Override the standard L^AT_EX list dimensions. Accept the same values as `\setlength`, including rubber lengths (e.g. `itemsep=4pt plus 2pt`). Each key may also take a comma-separated list of per-level values: e.g. `leftmargin={2.5em, 0em}`. The first value applies at level 1, the second at level 2, etc. Gaps in the list (e.g. `leftmargin={, 0em}`) cause the class default to be used for that level.

`labelindent`=⟨length / length list⟩ (no default)

Positions label left edges at ⟨length⟩ from the enclosing margin, adjusting `\labelsep` or `\itemindent` as needed. Positive values move rightward, negative leftward.

`tightspacing` (no value)

Sets all vertical spacing to the compact defaults that the standard document classes use for level-three lists (`topsep` and `itemsep` to 2pt with stretch/shrink, `parsep` to 0pt, `partopsep` to 1pt). Individual dimension keys set afterward override.

`nosep` (no value)

Sets `\topsep`, `\itemsep`, and `\parsep` all to zero.

8.2 Default types

`default type`=⟨type⟩ (initially default)

The type used when `\pitem` or `\ptag` is given a name or counter but no explicit type.

`default ptag type`=⟨type⟩ (initially empty)

If set, overrides `default type` for `\ptag` only.

level $n=\langle type \rangle$ (see below)

Default item type at nesting level n (1–5) when `\pitem` has no optional argument.

Level	Default type
1	numbered
2	leveltwo
3	levelthree
4	levelfour
5	levelfive

The `leveltwo`–`levelfive` types use special counters `numpropii`–`numpropv`, which reset automatically when `pitem` is used at lesser nesting levels. Other types can also use these counters.

8.3 Formatting and referencing equation numbers

equations (no value, **global only**)

Shares the `equation` counter between `\pitem[type=numbered]` and standard displayed equations. Also installs the equation format hooks (see below), so that `\oref` and `\nref` work with equation labels.

equation format $=\langle template \rangle$ (initially `(#1)`)

Shorthand: sets both `equation display format` and `equation ref format`. Also installs the equation format hooks independently of the `equations` option.

equation display format $=\langle template \rangle$ (initially `(#1)`)

Controls how equation tags appear in the PDF (via `\tagform@`). Use `#1` for the number. Does not affect `\ref` output. Also installs the equation format hooks independently of the `equations` option. Locally scoped.

equation ref format $=\langle template \rangle$ (initially `(#1)`)

Controls how `\ref` (and `\oref`, `\nref`) render equation labels. Use `#1` for the number. Does not affect the displayed tag. Also installs the equation format hooks independently of the `equations` option. Locally scoped.

9 Compatibility

The `propositions` package is designed to work with `hyperref`, `cleveref`, and `amsmath`. `amsmath` is required for `\ptag` and the `equations` option. With `cleveref` loaded, all `\pitem`s are assigned to a default `proposition` reference type; the `crefname` key can override this.

Recommended load order:

```
\usepackage{hyperref}
\usepackage{amsmath} % if using \ptag
\usepackage{propositions}
\usepackage{cleveref} % if used
```

10 Known issues

When using `\ptag` with a named counter (e.g. `\ptag[counter=P]`) inside an `ams-math` equation environment, `hyperref` may emit warnings of the form:

```
pdfTeX warning: destination with the same  
identifier (name{equation.N}) has been already  
used, duplicate ignored
```

These warnings are harmless and do not affect the correctness of cross-references.